M300

Install tools:

```
sudo apt update
sudo apt install docker-compose-v2
sudo apt install mysql-client-core-8.0
sudo apt install apache2
sudo apt install docker.io
sudo apt install openssh-client
sudo apt install rsync
sudo apt install python3
sudo apt install php
sudo apt install netcat-openbsd
sudo apt install telnet
sudo apt install wget
sudo apt install curl
sudo apt install dnsutils
sudo apt install apache2-utils
```

Terminals und Tools

SSH

```
vmadmin@vmLS3:~$ ssh-keyscan localhost | ssh-keygen -lf -
# localhost:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
# localhost:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
# localhost:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
# localhost:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
# localhost:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
3072 SHA256:NOJ4JCmvUMrQLTVC+xWqFGYmAMIKhiqxv7y+OJyK02U localhost (RSA)
256 SHA256:b8+4Gta0+Ucx4xoaRwF2MLWkeb3qJaPy24jF7IMYBb8 localhost (ECDSA)
256 SHA256:qgR0BFuXpMwRPvCdQgbxPPky7eiQmOwlvL2csEFzvEw localhost (ED25519)
vmadmin@vmLS3:~$
```

```
ssh-keyscan localhost | ssh-keygen -lf -    # Fingerprint des Host-Keys anzeigen
```

```
vmadmin@vmLP1:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/vmadmin/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vmadmin/.ssh/id_ed25519
Your public key has been saved in /home/vmadmin/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Ogew12UmZ+RLTqvHWR4fW0E2KqVNyfv30CtnF3pnhCM vmadmin@vmLP1.lan.m300.smartlearn.ch
The key's randomart image is:
+--[ED25519 256]--+
|          . .o.+ |
|         o  =o+ .|
|    .   . Xo o.. |
|     o . @ o..  .|
|    . o S + o oo.|
|     . o o +Eo+*+|
|      o o + ..++=|
|       o .   o ==|
|              =.o|
+----[SHA256]-----+
vmadmin@vmLP1:~$ ssh-copy-id 192.168.220.13
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vmadmin@192.168.220.13's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh '192.168.220.13'"
and check to make sure that only the key(s) you wanted were added.
```

```
ssh-keygen                                  # SSH-Schlüsselpaar erstellen (Standard)
ssh-keygen -t ecdsa                         # mit ECDSA-Algorithmus
```

```
vmadmin@vmLS3:~$ cat .ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIG5Ta9H7YY+Z/rYgQBQgFoRhKEsrvEtqICh7/rIzXwwg adm-remote@sladm-sr20.smartlearn.lan
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILhLrPB6dmPDtFftyEngpB9seLd8Q+psLtyog8EnckAz vmadmin@vmLP1.lan.m300.smartlearn.ch
```

```
ssh-copy-id 192.168.220.13                  # Public Key auf Server kopieren
```

```
vmadmin@vmLP1:~$ ssh vmadmin@192.168.220.13 "ip addr show"
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:56:01:24:3d brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname ens160
    inet 192.168.220.13/24 brd 192.168.220.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:fe01:243d/64 scope link
       valid_lft forever preferred_lft forever
```

```
ssh root@192.168.1.1                        # Login als root
ssh -l root 192.168.1.1                     # gleich, andere Schreibweise
ssh 192.168.220.13 "ip addr show"           # Remote-Befehl ohne Login-Shell
```

SSH Parameter:

```
-i    Identity-File definieren (privater Schlüssel)
-X    Graphische Oberfläche remote ansprechbar machen (X-Forwarding)
-F    Konfigurationsfile angeben
-o    einzelne Option anpassen
-N    kein Remote-Kommando ausführen
-T    kein Terminal verwenden
-p    Verbindung auf Portnummer aufbauen (statt Port 22)
-v    Debug-Ausgabe (auch mehrfach: -vvv)
```

IP-Adressen Referenz:

```
0.0.0.0       alle Interfaces
127.0.0.1     Localhost (nur lokal erreichbar)
::1           Localhost IPv6
192.168.1.1   statische Adresse eines Hosts
```

SCP

```
scp huhu.txt vmadmin@192.168.220.13:hehe                            # lokal → remote
scp vmadmin@192.168.220.13:hehe/huhu.txt huhu                       # remote → lokal
scp vmadmin@192.168.220.14:hihi.txt vmadmin@192.168.220.13:hihi     # remote → remote
scp -P 2222 archiv.zip vmadmin@127.0.0.1:~/                         # anderer Port
```

```
ssh -R 2222:127.0.0.1:22 vmadmin@192.168.220.13    # SCP über Reverse-Tunnel
```

SFTP

```
Connection to 192.168.220.13 closed.
vmadmin@vmLP1:~/scp$ sftp vmadmin@192.168.220.13
Connected to 192.168.220.13.
sftp> ls
test
sftp> ls -la
drwxr-x---    ? vmadmin  vmadmin      4096 Apr 26 13:22 .
drwxr-xr-x    ? root     root         4096 Jul  5  2024 ..
drwx------    ? vmadmin  vmadmin      4096 Jul  5  2024 .ansible
-rw-------    ? vmadmin  vmadmin        54 Apr 26 13:24 .bash_history
-rw-r--r--    ? vmadmin  vmadmin       220 Mar 31  2024 .bash_logout
-rw-r--r--    ? vmadmin  vmadmin      3818 Apr 26 13:22 .bashrc
drwx------    ? vmadmin  vmadmin      4096 Jul  5  2024 .cache
-rw-r--r--    ? vmadmin  vmadmin         0 Jul  5  2024 .hushlogin
-rw-------    ? vmadmin  vmadmin        20 Jan 26 15:38 .lesshst
-rw-r--r--    ? vmadmin  vmadmin       807 Mar 31  2024 .profile
drwx------    ? vmadmin  vmadmin      4096 Jul  5  2024 .ssh
-rw-r--r--    ? vmadmin  vmadmin         0 Jul  5  2024 .sudo_as_admin_successful
drwxrwxr-x    ? vmadmin  vmadmin      4096 Apr 26 13:23 test
```

```
sftp vmadmin@192.168.220.13    # Interaktiver Dateitransfer
```

RSYNC

```
rsync -rav --progress --delete iMrlvn.github.io vmadmin@192.168.220.13:website
# -r rekursiv, -a Metadaten erhalten, -v verbose
# --progress Fortschritt, --delete auf Ziel löschen was lokal nicht mehr existiert
```

```
rsync -avz datei.txt vmadmin@192.168.220.13:~/          # einzelne Datei kopieren
rsync -avz vmadmin@192.168.220.13:~/datei.txt ./        # Datei von Remote holen
```

SSH Config

```
  GNU nano 7.2                                                    /root/.ssh/config
Host remote
  HostName 192.168.220.13
  User vmadmin
  IdentityFile ~/.ssh/id_ed25519
  IdentitiesOnly yes
```

```
root@vmLP1:/home/vmadmin# nano ~/.ssh/config
root@vmLP1:/home/vmadmin# ssh remote
The authenticity of host '192.168.220.13 (192.168.220.13)' can't be established.
ED25519 key fingerprint is SHA256:qgR0BFuXpMwRPvCdQgbxPPky7eiQmOwlvL2csEFzvEw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Alias in ~/.ssh/config:

```
Host meinserver
  HostName 192.168.220.13
  User vmadmin
  IdentityFile ~/.ssh/key
  IdentitiesOnly yes
```

SSH Tunneling

MYSQL Container

```
vmadmin@vmLS3:~$ sudo docker run -d \
  --name my-mysql \
  -e MYSQL_ROOT_PASSWORD=my-secret-pw \
  -p 127.0.0.1:3306:3306 \
  mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
bb5107df7baa: Pull complete
191b09bcaa2f: Pull complete
85d9f281a64d: Download complete
```

```
docker run -d \
  --name my-mysql \
  -e MYSQL_ROOT_PASSWORD=my-secret-pw \
  -p 127.0.0.1:3306:3306 \
  mysql:latest
```

sudo apt update && sudo apt install mysql-client-core-8.0

```
vmadmin@vmLS3:~$ mysql -h 127.0.0.1 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 9.7.0 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database LS3LOKAL;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> create database LS3LOKAL;
```

```
ss -tlpn
```

Output:

```
LISTEN 0 4096 127.0.0.1:3306 0.0.0.0:*
```

```
mysql -h 127.0.0.1 -u root -p    # Verbinden
create database LS3LOKAL;         # DB erstellen
```

Tunneling

```
ssh -L 127.0.0.1:13306:127.0.0.1:3306 vmadmin@192.168.220.13        # mit Login-Shell
```

```
vmadmin@vmLP1:~$ mysql -h 127.0.0.1 -P 13306 -u root  -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 9.7.0 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

```
vmadmin@vmLP1:~$ ssh -fN -L 127.0.0.1:13306:127.0.0.1:3306  vmadmin@192.168.220.13
vmadmin@vmLP1:~$ ps aux | grep ssh
vmadmin     1816  0.0  0.1 162652  7040 ?        Ssl  13:12   0:00 /usr/libexec/gcr-ssh-agent --base-dir /run/user/1000/gcr
vmadmin     3921  0.0  0.0   8432  5888 ?        S    13:15   0:00 /usr/bin/ssh-agent -D -a /run/user/1000/keyring/.ssh
root        5259  0.0  0.1  15328  8448 pts/2    S+   13:39   0:00 ssh remote
vmadmin     6725  0.0  0.0  15328  5156 ?        Ss   14:08   0:00 ssh -fN -L 127.0.0.1:13306:127.0.0.1:3306 vmadmin@192.168.220.13
vmadmin     6730  0.0  0.0   7276  2304 pts/1    S+   14:08   0:00 grep --color=auto ssh
vmadmin@vmLP1:~$ kill 6725
```

```
ssh -L 127.0.0.1:13306:127.0.0.1:3306 -N -f vmadmin@192.168.220.13  # ohne Shell, im
Hintergrund
```

```
mysql -h 127.0.0.1 -P 13306 -u root -pmy-secret-pw    # von aussen verbinden
```

```
ps aux | grep ssh                # laufende SSH-Prozesse anzeigen
kill <PID>                       # Prozess beenden
pkill -f "ssh -L 127.0.0.1:13306"  # Tunnel gezielt beenden
```

SSH durch FW (Reverse Tunnel)

```
vmadmin@vmLP1:~$ ssh -fN -R 127.0.0.1:2222:127.0.0.1:22 vmadmin@192.168.220.13
vmadmin@vmLP1:~$
```

```
vmadmin@vmLS3:~$ ssh -p 2222 localhost
vmadmin@localhost's password:
```

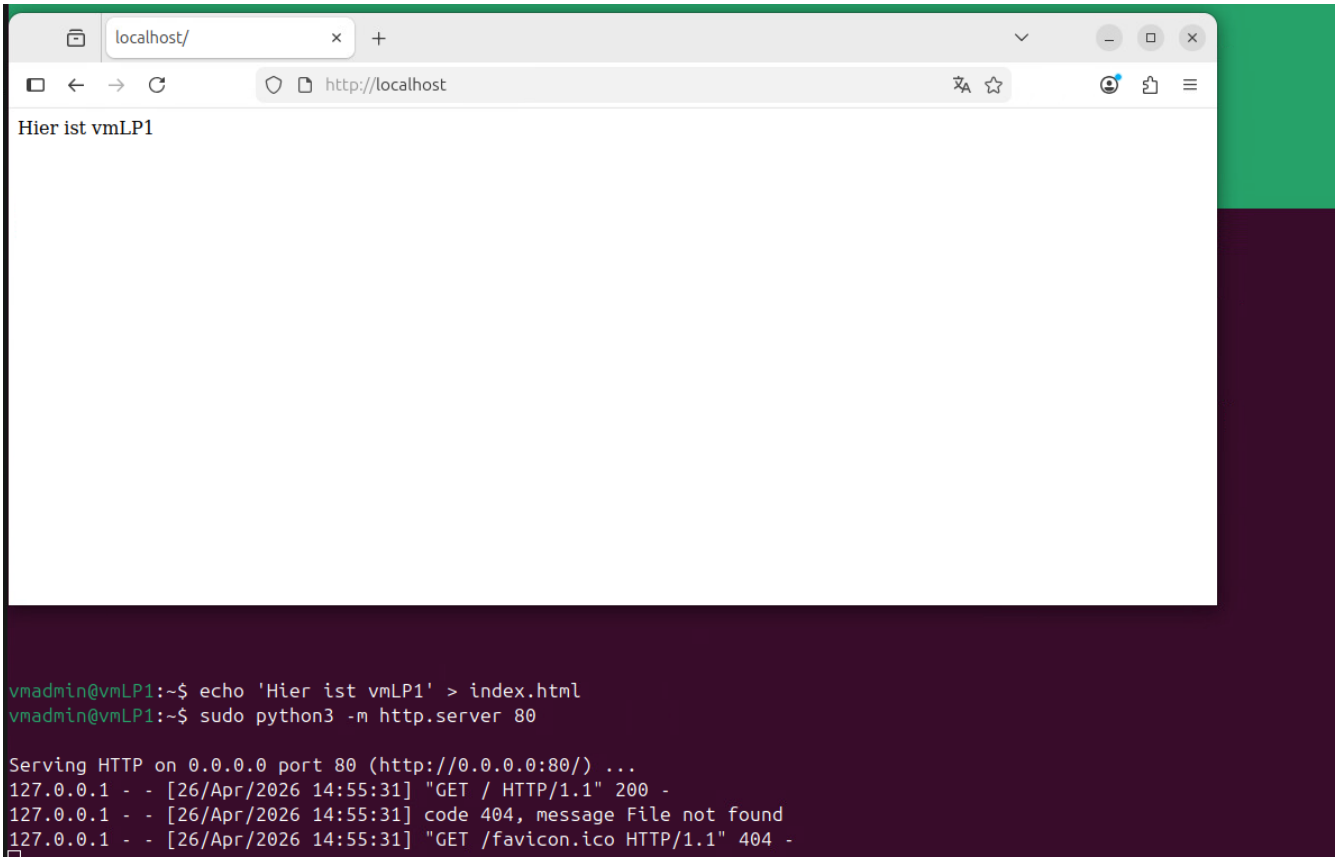
Szenario: Server → Client gesperrt durch Firewall

```
# Auf Client ausführen:
ssh -R 12222:127.0.0.1:22 vmadmin@192.168.220.13
```

```
# Danach vom Server auf Client verbinden:
ssh -p 12222 vmadmin@127.0.0.1
```

```
pkill -f "ssh -R 12222"    # Reverse-Tunnel beenden
```

Webserver hinter NAT




```
# Auf Client (vmLP1):
echo 'Hier ist vmLP1' > index.html
sudo python3 -m http.server 80
ssh -R 0.0.0.0:8123:127.0.0.1:80 vmadmin@192.168.220.13
```

```
# Auf Server (vmLS3) testen:
wget 127.0.0.1:8123
```

Problem beim WGET auf VMLS3

```
vmadmin@vmLS3:~$ sudo sed -i 's/#GatewayPorts no/GatewayPorts yes/' /etc/ssh/sshd_config
vmadmin@vmLS3:~$ sudo systemctl restart ssh.socket
sudo systemctl restart ssh.service
vmadmin@vmLS3:~$
```

GatewayPorts aktivieren (damit Tunnel auf 0.0.0.0 lauscht):

```
# In /etc/ssh/sshd_config auf dem Server:
GatewayPorts yes
```

```
sudo systemctl restart ssh
```

Falls Prozess 8123 immer noch auf 127...zeigt das hier machen:

```
vmadmin@vmLS3:~$ ss -tlpn | grep 8123
LISTEN 0      128        127.0.0.1:8123       0.0.0.0:*
LISTEN 0      128            [::1]:8123          [::]:*
vmadmin@vmLS3:~$ sudo ss -tlpn | grep 8123
LISTEN 0      128        127.0.0.1:8123       0.0.0.0:*    users:(("sshd",pid=6408,fd=8))
LISTEN 0      128            [::1]:8123          [::]:*    users:(("sshd",pid=6408,fd=7))
vmadmin@vmLS3:~$ sudo kill 6408
vmadmin@vmLS3:~$ ss -tlpn | grep 8123
vmadmin@vmLS3:~$ ss -tlpn | grep 8123
LISTEN 0      128          0.0.0.0:8123       0.0.0.0:*
LISTEN 0      128             [::]:8123          [::]:*
```

Danach sollte es auf 0.0.0.0 zeigen.

TUNNEL NEU AUFBAUEN!!

```
vmadmin@vmLP1:~$ pkill -f "ssh -fN -R"
vmadmin@vmLP1:~$ ssh -fN -R 0.0.0.0:8123:127.0.0.1:80 vmadmin@192.168.220.13
vmadmin@vmLP1:~$
```

Erneut versuchen:

```
                                        IP: 192.168.210.31/24
                                       MAC: 00:50:56:00:24:1e

                                    Domain: lan.m300.smartlearn.ch
                                  Hostname:              vmLP1
                                   Gateway:      192.168.210.1
                                       DNS:      192.168.210.1

vmadmin@vmLP1:~$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
127.0.0.1 - - [26/Apr/2026 15:20:40] "GET / HTTP/1.1" 200 -
```

```
vmadmin@vmLS3:~$ wget 127.0.0.1:8123
--2026-04-26 15:20:40--  http://127.0.0.1:8123/
Connecting to 127.0.0.1:8123... connected.
HTTP request sent, awaiting response... 200 OK
Length: 15 [text/html]
Saving to: ‘index.html.1’

index.html.1        100%[===================>]      15  --.-KB/s    in 0s

2026-04-26 15:20:40 (2.48 MB/s) - ‘index.html.1’ saved [15/15]

vmadmin@vmLS3:~$
```

Tools

SSH

Wieder SSH einrichten

```
ssh-keygen -t ecdsa
```

```
ssh-copy-id 192.168.220.13
```

```
ssh 192.168.220.13
```

Debugging SSH:

```
ssh -v vmadmin@192.168.220.13      # verbose, zeigt wo es hängt
ssh -vvv vmadmin@192.168.220.13    # maximales Detail
```

auf vmLS3 mit Root:

```
root@vmLS3:~# tail -n 1 ~vmadmin/.ssh/authorized_keys >> .ssh/authorized_keys
root@vmLS3:~#
logout
vmadmin@vmLS3:~$
logout
Connection to 192.168.220.13 closed.
vmadmin@vmLP1:~$ ssh root@192.168.220.13
```

Disk

```
root@vmLS3:~# fdisk /dev/sdd

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xcfc5bfa7.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-67108863, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-67108863, default 67108863):

Created a new partition 1 of type 'Linux' and of size 32 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
lsblk                                          # Disks und Partitionen anzeigen
```

```
sudo fdisk /dev/sdd                            # Disk partitionieren
# → n  (neue Partition), Enter, Enter, Enter
# → w  (schreiben und beenden)
```

Nun formatieren wir die Partition /dev/sdd1 mit dem Dateisystem ext4 und geben ihn das Label m300data.

```
root@vmLS3:~# mkfs -t ext4 -L m300data /dev/sdd1
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 8388352 4k blocks and 2097152 inodes
Filesystem UUID: 08b6e0ce-7c24-4c47-9703-01eeb09dc959
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

root@vmLS3:~# blkid
/dev/sr0: BLOCK_SIZE="2048" UUID="2222-01-01-12-34-56-78" LABEL="SML_KS" TYPE="iso9660"
/dev/sda2: UUID="dbac683c-0f3d-4c6a-9412-7f7c2ca5bcbb" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="49015875-bc25-4c2c-9ca7-f21a46076953"
/dev/loop1: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/sdd1: LABEL="m300data" UUID="08b6e0ce-7c24-4c47-9703-01eeb09dc959" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="cfc5bfa7-01"
/dev/loop2: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop0: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/sda1: PARTUUID="e4ce965c-7f08-439f-b750-87f779f4fe4d"
```

```
sudo mkfs -t ext4 -L m300data /dev/sdd1        # Partition formatieren
blkid                                           # UUIDs und Dateisysteme anzeigen
```

```
sudo mkdir /m300data                                # Mountpoint erstellen
```

Automatisch mounten bei Reboot

sudo nano /etc/fstab

```
  GNU nano 7.2                                                          /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
# / was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/dbac683c-0f3d-4c6a-9412-7f7c2ca5bcbb / ext4 defaults 0 1
LABEL=m300data     /m300data    ext4   defaults 0 0
```

nun mounten und prüfen

```
root@vmLS3:~# mount -a
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
root@vmLS3:~# mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=458976k,nr_inodes=114744,mode=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=98476k,mode=755,inode64)
/dev/sda2 on / type ext4 (rw,relatime)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
tmpfs on /run/lock type tmpfs (rw,nosuid,nodev,noexec,relatime,size=5120k,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=32,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=3552)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=98472k,nr_inodes=24618,mode=700,uid=1000,gid=1000,inode64)
/var/lib/snapd/snaps/snapd_26865.snap on /snap/snapd/26865 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide,x-gvfs-hide)
/var/lib/snapd/snaps/core24_1587.snap on /snap/core24/1587 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide,x-gvfs-hide)
/var/lib/snapd/snaps/docker_3377.snap on /snap/docker/3377 type squashfs (ro,nodev,relatime,errors=continue,threads=single,x-gdu.hide,x-gvfs-hide)
tmpfs on /run/snapd/ns type tmpfs (rw,nosuid,nodev,noexec,relatime,size=98476k,mode=755,inode64)
nsfs on /run/snapd/ns/docker.mnt type nsfs (rw)
tmpfs on /run/user/0 type tmpfs (rw,nosuid,nodev,relatime,size=98472k,nr_inodes=24618,mode=700,inode64)
/dev/sdd1 on /m300data type ext4 (rw,relatime)
root@vmLS3:~# df
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             98476    1316     97160   2% /run
/dev/sda2       8170952 3488140   4289184  45% /
tmpfs            492376       0    492376   0% /dev/shm
tmpfs              5120       0      5120   0% /run/lock
tmpfs             98472      12     98460   1% /run/user/1000
```

```
mount -a                                            # Alle fstab-Einträge mounten
lsblk                                               # Mount prüfen
```

Docker

```
root@vmLS3:~# mkdir /m300data/docker

root@vmLS3:~# ln -s /m300data/docker /var/lib/docker
root@vmLS3:~# apt install -y docker.io docker-compose-v2
```

```
mkdir /m300data/docker
ln -s /m300data/docker /var/lib/docker          # Docker-Daten auf separate Disk
sudo apt install -y docker.io docker-compose-v2
docker run hello-world                           # Installation testen
```

vmadmin zu Docker Gruppe hinzufügen:

```
sudo usermod -aG docker vmadmin
```

Docker Befehle:

```
docker ps                        # laufende Container
docker ps -a                     # alle Container (auch gestoppte)
docker images                    # heruntergeladene Images
docker stop <name>               # Container stoppen
docker rm <name>                 # Container löschen
docker logs <containername>      # Logs anzeigen
docker inspect <containername>   # Details & Netzwerk-Config
docker exec -it <name> bash      # Shell in Container öffnen
```

Volumes:

```
docker volume create portainer_data    # Volume erstellen
docker volume ls                       # Volumes anzeigen
docker volume rm <name>               # Volume löschen
```

Docker Compose:

```
docker compose up              # starten (Vordergrund)
docker compose up -d           # starten (Hintergrund)
docker compose down            # stoppen und entfernen
docker compose ps              # Status anzeigen
docker compose logs            # Logs anzeigen
```

Portainer

Volume erstellen

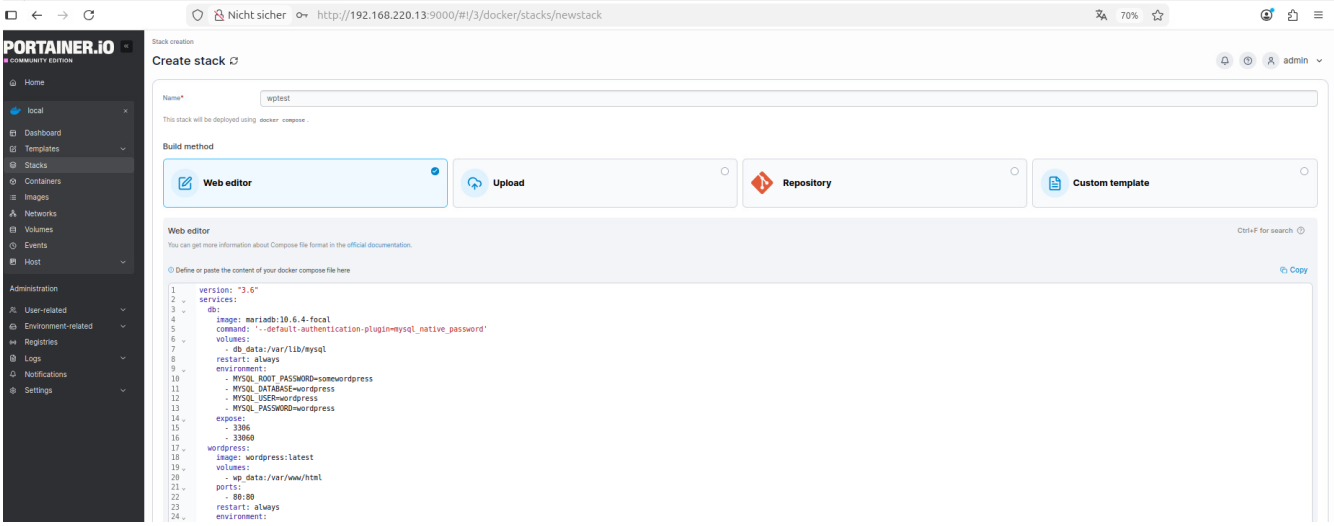
```
root@vmLS3:~# docker volume create portainer_data
portainer_data
root@vmLS3:~# docker volume ls
DRIVER    VOLUME NAME
local     portainer_data
```

Portainer einrichten:

```
docker run -d -p 9000:9000 --name portainer --restart always \
    -v /var/run/docker.sock:/var/run/docker.sock \
    -v portainer_data:/data \
    portainer/portainer-ce:lts
```

nun http://192.168.220.13:9000 im Browser

Docker Compose



Siehe docker-compose.yml;

```
version: "3.6"
    services:
        db:
            image: mariadb:10.6.4-focal
            command: '--default-authentication-plugin=mysql_native_password'
            volumes:
                - db_data:/var/lib/mysql
            restart: always
            environment:
                - MYSQL_ROOT_PASSWORD=somewordpress
                - MYSQL_DATABASE=wordpress
                - MYSQL_USER=wordpress
                - MYSQL_PASSWORD=wordpress
            expose:
                - 3306
                - 33060
        wordpress:
            image: wordpress:latest
            volumes:
                - wp_data:/var/www/html
            ports:
                - 80:80
```

```
            restart: always
            environment:
                - WORDPRESS_DB_HOST=db
                - WORDPRESS_DB_USER=wordpress
                - WORDPRESS_DB_PASSWORD=wordpress
                - WORDPRESS_DB_NAME=wordpress
    volumes:
        db_data:
        wp_data:
```

Danach: http://vmls3.m300.smartlearn.ch/wp-admin/install.php

`python3 -m http.server 1234`

`php -S localhost:8080 php -S 0.0.0.0:8080`

`while true; do cat index.http | nc -l 8000; done`

Siehe index.http dafür

docker run -d -p 1234:80 traefik/whoami wget -qO- http://192.168.220.13:1234

telnet www.gibb.ch 80

wget http://gibb.ch

curl http://gibb.ch

Apache2

`sudo apt install apache2`

Apache aktivieren: image::27EnableApache.png[]

```
vmadmin@vmLS3:~$ sudo mkdir -p /var/www/html/myweb
vmadmin@vmLS3:~$ sudo chown -R www-data:www-data /var/www/html/myweb
vmadmin@vmLS3:~$ echo '<h1>Hallo von Apache!</h1>' | sudo tee /var/www/html/myweb/index.html
<h1>Hallo von Apache!</h1>
```

VirtualHost konfigurieren:

sudo nano /etc/apache2/sites-available/myweb.conf

```
<VirtualHost *:80>
    ServerName myweb.local
    DocumentRoot /var/www/html/myweb
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

```
sudo a2ensite myweb
sudo a2dissite 000-default    # Standard-Site deaktivieren
sudo systemctl reload apache2
```

Testen

```
wget -qO- http://192.168.220.13
```

Debugging

```
sudo systemctl status apache2      # Status prüfen
sudo apache2ctl -S                 # VirtualHost-Konfiguration anzeigen
sudo ss -tlpn | grep :80          # Port 80 prüfen
sudo journalctl -xeu apache2      # Logs bei Fehler
```

DNS

Wichtigste DNS Record-Typen:

Record	Bedeutung
A	Hostname → IPv4-Adresse
CNAME	Alias → anderer Hostname (z.B. www → vmls3.lan.m300.smartlearn.ch)
MX	Mail-Server für eine Domain
NS	Nameserver der Zone
SOA	Start of Authority – Zonen-Metadaten (Serial, TTL, etc.)
TXT	Freitext – oft für SPF, DKIM, Domain-Verifizierung

ACHTUNG: Zuerst alle Images pullen:

```
docker pull eafxx/bind
```

```
vmadmin@vmLS3:~$ systemctl stop systemd-resolved.service
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to stop 'systemd-resolved.service'.
Authenticating as: vmadmin
Password:
==== AUTHENTICATION COMPLETE ====
vmadmin@vmLS3:~$ systemctl disable systemd-resolved.service
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: vmadmin
Password:
==== AUTHENTICATION COMPLETE ====
Removed "/etc/systemd/system/dbus-org.freedesktop.resolve1.service".
Removed "/etc/systemd/system/sysinit.target.wants/systemd-resolved.service".
==== AUTHENTICATING FOR org.freedesktop.systemd1.reload-daemon ====
Authentication is required to reload the systemd state.
Authenticating as: vmadmin
Password:
==== AUTHENTICATION COMPLETE ====
```

Somit ist Port 53 freigegeben

DNS als Container

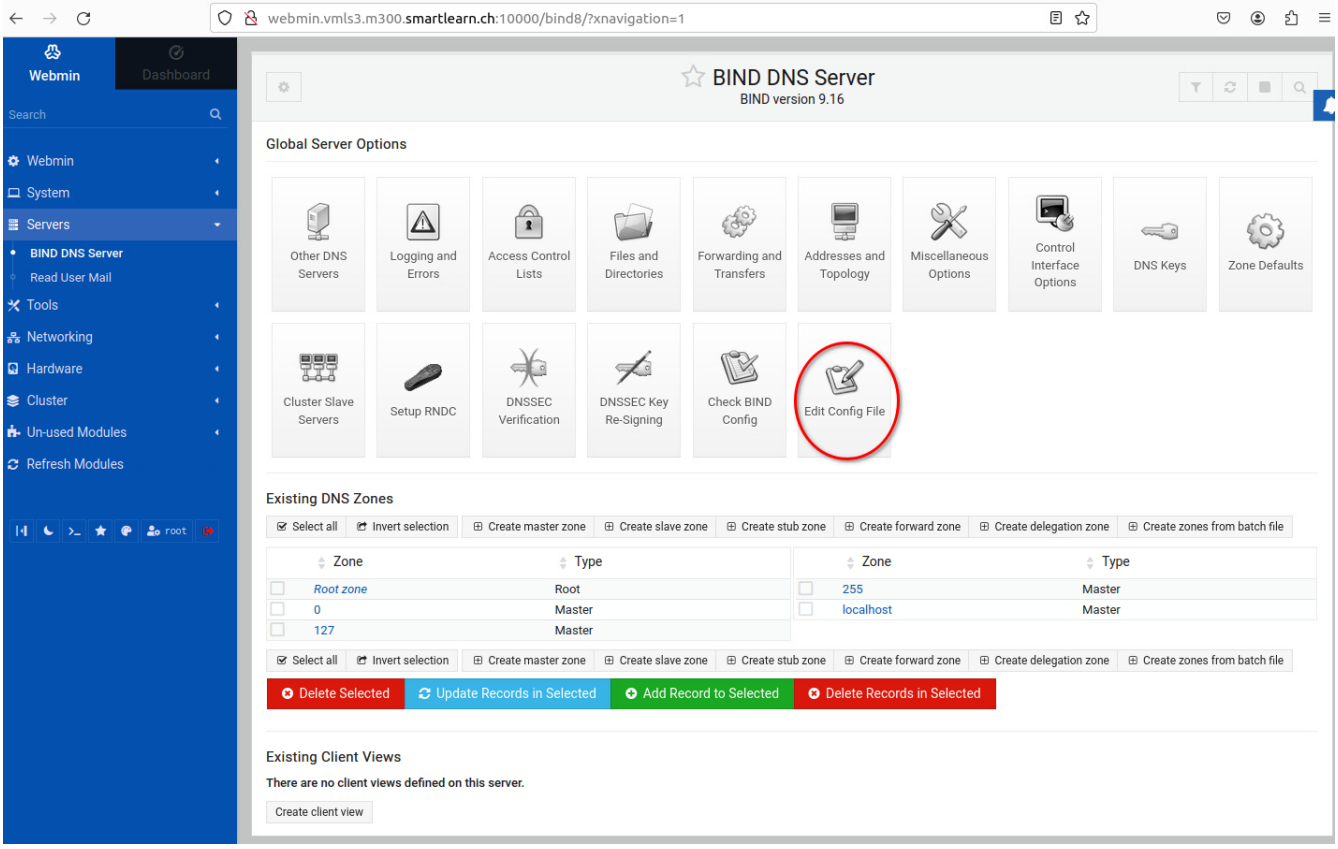
```
version: '3.6'
services:
    bind:
        image: eafxx/bind
        restart: unless-stopped
        ports:
            - 53:53/tcp
            - 53:53/udp
            - 10000:10000
        expose:
            - 10000
        volumes:
            - 'bind:/data'
            - './named.conf.options:/etc/bind/named.conf.options'
        environment:
            - WEBMIN_ENABLED=true
            - WEBMIN_INIT_SSL_ENABLED=false
            - WEBMIN_INIT_REFERERS=vmls3.m300.smartlearn.ch
            - WEBMIN_INIT_REDIRECT_PORT=10000
            - ROOT_PASSWORD=sml12345
            - TZ=Europe/Zurich
volumes:
    bind:
```

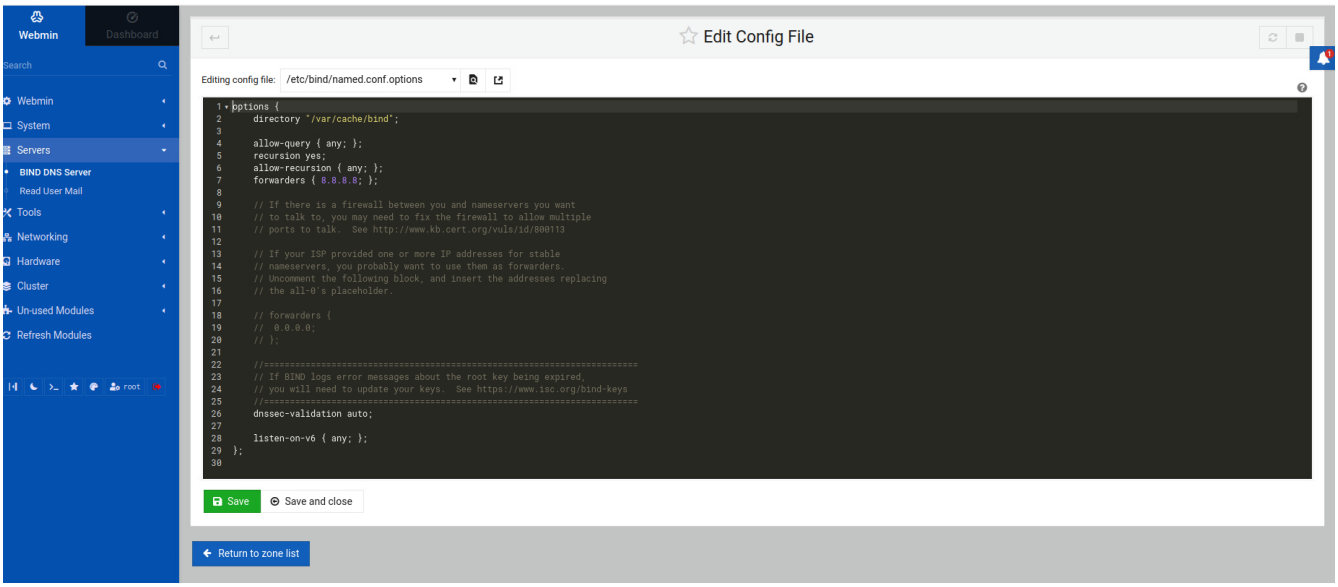
Output:

```
vmadmin@vmLS3:~$ docker ps -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED              STATUS                      PORTS
                                                                   NAMES
79a1e77fd222   eafxx/bind                 "/sbin/entrypoint.sh…"   About a minute ago   Up 52 seconds               0.0.0.0:53->53/tcp, [::]:53->53/tcp, 0.0.0.0:10000->10000/tcp, 0.0
.0.0:53->53/udp, [::]:10000->10000/tcp, [::]:53->53/udp   dns-webmin-bind-1
b434c5eb2c68   traefik/whoami             "/whoami"                2 hours ago          Exited (2) 44 minutes ago
                                                                   gracious_keller
17ab7d0192df   portainer/portainer-ce:lts "/portainer"             2 hours ago          Up 3 minutes                8000/tcp, 9443/tcp, 0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp
                                                                   portainer
8c7ec96ec403   mysql:latest               "docker-entrypoint.s…"   6 hours ago          Exited (255) 2 hours ago    127.0.0.1:3306->3306/tcp, 33060/tcp
                                                                   my-mysql
vmadmin@vmLS3:~$ sudo ss -tlpn | grep 10000
LISTEN 0      4096         0.0.0.0:10000      0.0.0.0:*    users:(("docker-proxy",pid=11212,fd=7))
LISTEN 0      4096            [::]:10000         [::]:*    users:(("docker-proxy",pid=11220,fd=7))
```

Dann: http://192.168.220.13:10000/

Mit URL





Auf vmLP1:

sudo cat /etc/resolv.conf

nameserver anpassen:

```
nameserver 192.168.220.13
options edns0 trust-ad
search .
```

test: nslookup webmin.vmls3.m300.smartlearn.ch

Debugging DNS:

```
nslookup test.lan.m300.smartlearn.ch 192.168.220.13    # einfacher Test
dig test.lan.m300.smartlearn.ch @192.168.220.13        # detaillierter Test
dig -x 192.168.220.13 @192.168.220.13                  # Reverse-Lookup testen
```

Webmin

Editing config file: /etc/bind/named.conf.options

```
options {
    directory "/var/cache/bind";

    forwarders {
        8.8.8.8;
    };
    allow-recursion {
        192.168.210.0/24;
        192.168.220.0/24;
    };
    dnssec-validation no;
    listen-on-v6 { };
    auth-nxdomain no;
    empty-zones-enable no;
};

```

```
sudo docker exec dns-webmin-bind-1 bash -c 'cat > /etc/bind/named.conf.local << EOF
// lokale zonen
//
zone "lan.m300.smartlearn.ch" {
    type master;
    file "/etc/bind/db.lan.m300.smartlearn.ch";
};
zone "220.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.220";
```

```
};
zone "210.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.210";
};
EOF'
```

Zonen Records anpassen:

```
sudo docker exec dns-webmin-bind-1 bash -c 'cat > /etc/bind/db.192.168.220 << EOF
\$TTL 604800
@   IN  SOA vmls3.lan.m300.smartlearn.ch. admin.lan.m300.smartlearn.ch. (
            2026042601 ; Serial
            604800     ; Refresh
            86400      ; Retry
            2419200    ; Expire
            604800 )   ; Negative Cache TTL
@   IN  NS  vmls3.lan.m300.smartlearn.ch.
13  IN  PTR vmls3.lan.m300.smartlearn.ch.
EOF'
```

```
sudo docker exec dns-webmin-bind-1 bash -c 'cat > /etc/bind/db.192.168.210 << EOF
\$TTL 86400
@   IN SOA ns.m300.smartlearn.ch. root.m300.smartlearn.ch. (
                        2026042601
                        14400
                        1800
                        1209600
                        3600 )
@   172800  IN NS ns.m300.smartlearn.ch.
1   IN PTR vmlf1.lan.m300.smartlearn.ch.
31  IN PTR vmlp1.lan.m300.smartlearn.ch.
EOF'
```

```
sudo docker exec dns-webmin-bind-1 bash -c 'cat > /etc/bind/db.lan.m300.smartlearn.ch
<< EOF
\$TTL 3600
lan.m300.smartlearn.ch. IN SOA ns.m300.smartlearn.ch. root.m300.smartlearn.ch. (
                        2026042601
                        3600
                        600
                        1209600
                        3600 )
lan.m300.smartlearn.ch. IN NS ns.m300.smartlearn.ch.
vmls3   IN A 192.168.220.13
vmlp1   IN A 192.168.210.31
test    IN A 3.4.5.6
```

```
EOF'
```

NEUSTARTEN:

```
sudo docker exec dns-webmin-bind-1 rndc reload
systemctl restart named
```

Test:

```
vmadmin@vmLS3:~$ sudo docker exec dns-webmin-bind-1 cat /etc/bind/db.192.168.220
$TTL 604800
@   IN  SOA vmls3.lan.m300.smartlearn.ch. admin.lan.m300.smartlearn.ch. (
            2026042601 ; Serial
            604800     ; Refresh
            86400      ; Retry
            2419200    ; Expire
            604800 )   ; Negative Cache TTL
@   IN  NS  vmls3.lan.m300.smartlearn.ch.
13  IN  PTR vmls3.lan.m300.smartlearn.ch.
vmadmin@vmLS3:~$ nslookup 192.168.210.1 192.168.220.13
1.210.168.192.in-addr.arpa      name = vmlf1.lan.m300.smartlearn.ch.
```

```
nslookup test.lan.m300.smartlearn.ch 192.168.220.13
Server:     192.168.220.13
Address:    192.168.220.13#53
Name:   test.lan.m300.smartlearn.ch
Address: 3.4.5.6
```

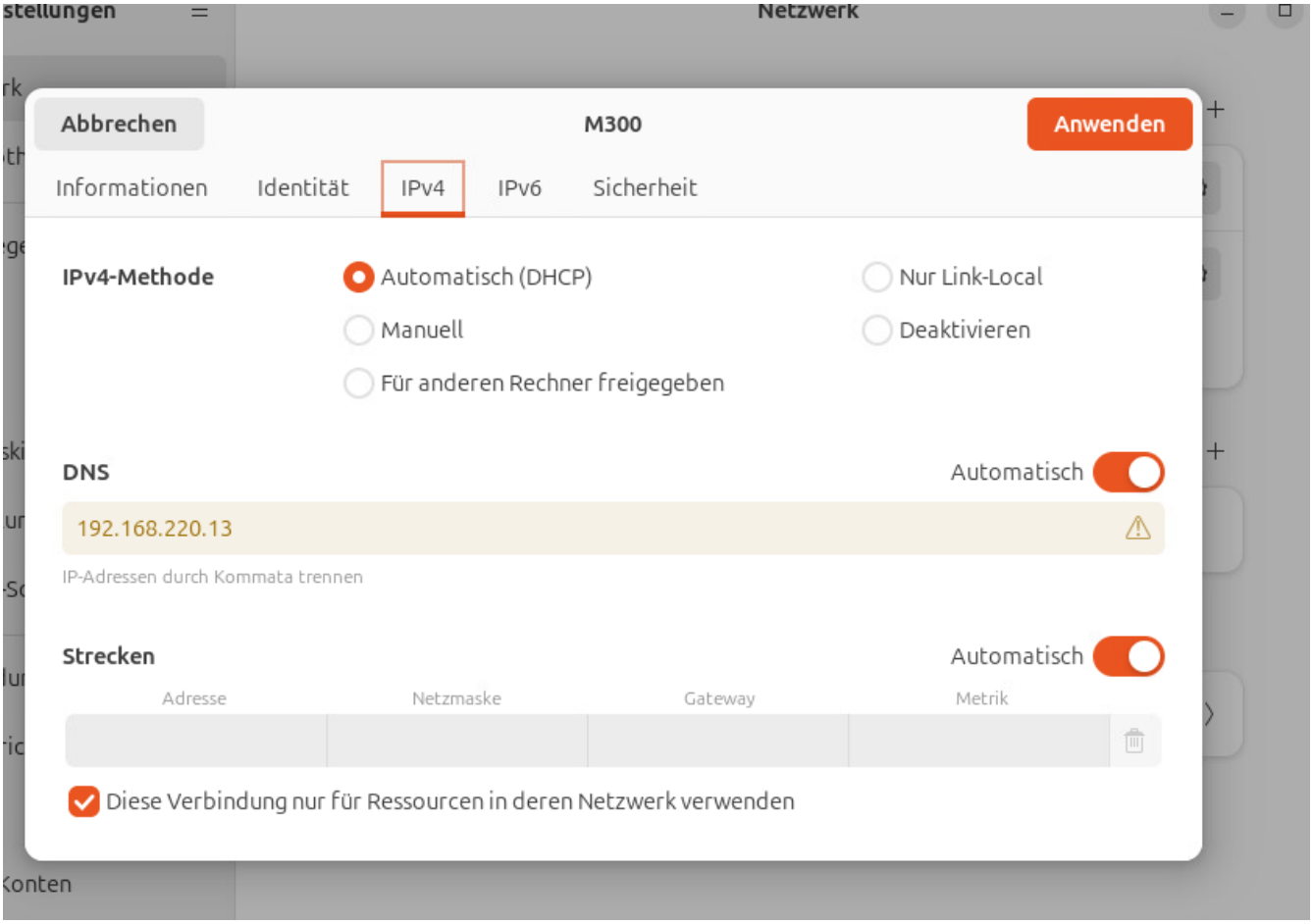
Neue Record hinzufügen:

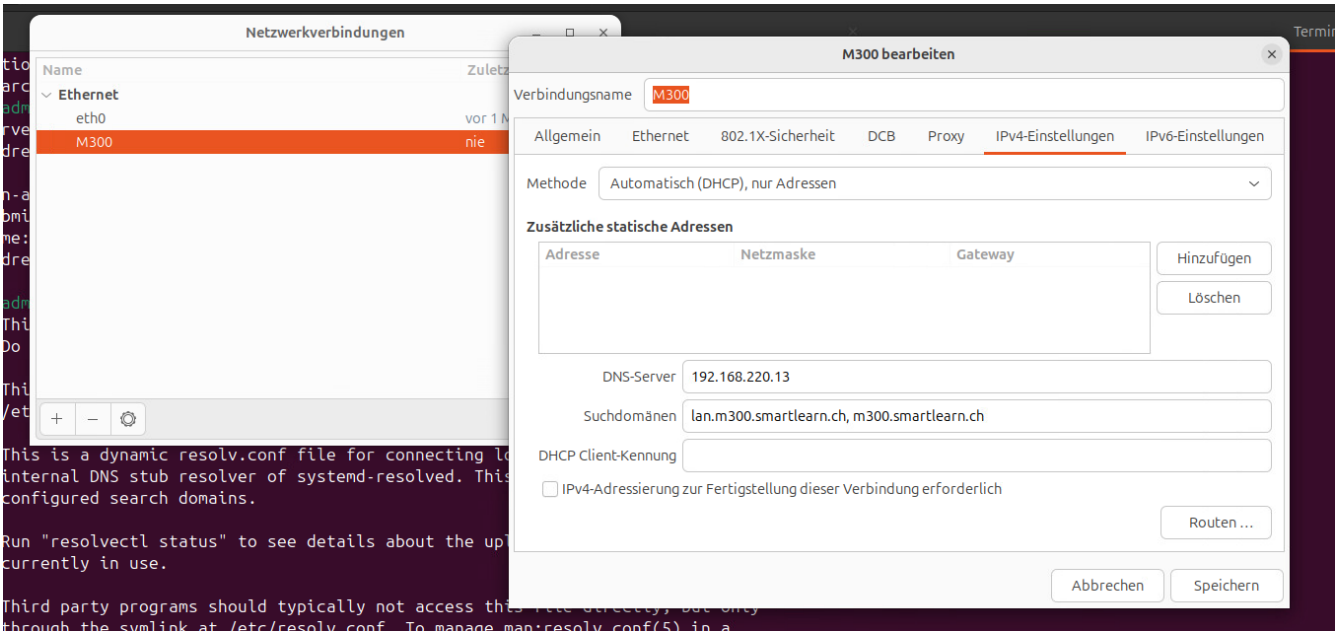
```
sudo docker exec dns-webmin-bind-1 bash -c 'echo "vmls5   IN A 192.168.220.15" >>
/etc/bind/db.lan.m300.smartlearn.ch'
sudo docker exec dns-webmin-bind-1 rndc reload
```

```
nslookup test.lan.m300.smartlearn.ch 192.168.220.13
Server:     192.168.220.13
Address:    192.168.220.13#53
Name:   test.lan.m300.smartlearn.ch
Address: 3.4.5.6
```

Wichtige Befehle:

```
sudo docker exec dns-webmin-bind-1 cat /etc/bind/db.192.168.220
```





WICHTIG!

```
sudo cat /etc/systemd/resolved.conf | grep DNSSEC
DNSSEC=no
```

`resolvectl status`

```
Global
```

```
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub
Link 2 (eth0)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.220.13
    DNS Servers: 192.168.220.13
```

Suchdomänen:

```
sudo resolvectl domain eth0 lan.m300.smartlearn.ch m300.smartlearn.ch
```

```
vmadmin@vmLP1:~$ resolvectl status
Global
         Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (eth0)
    Current Scopes: DNS
         Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.220.13
       DNS Servers: 192.168.220.13
        DNS Domain: lan.m300.smartlearn.ch m300.smartlearn.ch
vmadmin@vmLP1:~$ ping vmlf1
PING vmlf1.m300.smartlearn.ch (192.168.220.1) 56(84) bytes of data.
64 bytes from 192.168.220.1: icmp_seq=1 ttl=64 time=0.130 ms
64 bytes from 192.168.220.1: icmp_seq=2 ttl=64 time=0.255 ms
64 bytes from 192.168.220.1: icmp_seq=3 ttl=64 time=0.251 ms
64 bytes from 192.168.220.1: icmp_seq=4 ttl=64 time=0.216 ms
64 bytes from 192.168.220.1: icmp_seq=5 ttl=64 time=0.193 ms
^C
--- vmlf1.m300.smartlearn.ch ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4105ms
rtt min/avg/max/mdev = 0.130/0.209/0.255/0.045 ms
```

Traefik

Zuerst:

```
docker network create traefik-proxy
apt install apache2-utils
htpasswd -cB pfwile admin
```

```
root@vmLS5:/home/vmadmin/traefik# htpasswd -cB pwfile admin
New password:
Re-type new password:
Adding password for user admin
```

Dann diesen Struktur aufbauen.

Siehe traefik Ordner

```
vmadmin@vmLS3:~$ find /home/vmadmin/traefik -type f -o -type d | sort
/home/vmadmin/traefik
/home/vmadmin/traefik/data
/home/vmadmin/traefik/data/letsencrypt
/home/vmadmin/traefik/docker-compose.yml
/home/vmadmin/traefik/.env
/home/vmadmin/traefik/pwfile
/home/vmadmin/traefik/traefik
/home/vmadmin/traefik/traefik/certs
/home/vmadmin/traefik/traefik/certs/default
/home/vmadmin/traefik/traefik/certs/default/cert.pem
/home/vmadmin/traefik/traefik/certs/default/privkey.pem
/home/vmadmin/traefik/traefik/dynamic.yml
/home/vmadmin/traefik/traefik/traefik.yml
```

anschliessend:

```
docker compose up -d
[+] Running 1/1
✔ Container traefik  Started
```

80 und 443 müssen freigegeben werden: 0.0.0.0:80→80/tcp 0.0.0.0:443→443/tcp

sonst freigeben mit: sudo ufw allow 80/tcp sudo ufw allow 443/tcp

alternative: sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

Beispiel Anwendung

simples nginx docker-compose: services: nginx: image: nginx:1.20 labels: - "traefik.enable=true" - "traefik.http.routers.nginx.entrypoints=websecure" - "traefik.http.routers.nginx.rule=Host(nginx.m300.smartlearn.ch)" - "traefik.http.routers.nginx.tls=true" - "traefik.http.services.nginx.loadbalancer.server.port=80" networks: - traefik-proxy networks: traefik-proxy: external: true

Apache PHP Seite: services: nginx: image: php:8.1-apache labels: - "traefik.enable=true" - "traefik.http.routers.php.entrypoints=websecure" - "traefik.http.routers.php.rule=Host(php.m300.smartlearn.ch)" - "traefik.http.routers.php.tls=true" - "traefik.http.services.php.loadbalancer.server.port=80" networks: - traefik-proxy volumes: - ./html:/var/www/html networks: traefik-proxy: external: true

html seite noch erfassen: mkdir html echo '<?php phpinfo();' | sudo tee html/index.php

Mit Middleware (PathPrefix /test mit Strip):

```
services:
  nginx:
    image: php:8.1-apache
```

```
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.php.entrypoints=websecure"
      - "traefik.http.routers.php.rule=Host(`php.m300.smartlearn.ch`) &&
PathPrefix(`/test`)"
      - "traefik.http.routers.php.tls=true"
      - "traefik.http.routers.php.middlewares=strip-testpath@docker" # aktiviert
Middleware strip für diesen Container
      - "traefik.http.services.php.loadbalancer.server.port=80"
      - "traefik.http.middlewares.strip-testpath.stripprefix.prefixes=/test" #
entfernt /test bei Aufruf
    networks:
      - traefik-proxy
    volumes:
      - ./html:/var/www/html
networks:
  traefik-proxy:
    external: true
```

Unterseite noch erstellen:

```
mkdir -p html/test
echo '<?php echo "<h1>Das ist die Test-Seite</h1>";' | sudo tee html/test/index.php
```

Erreichbar unter: https://php.m300.smartlearn.ch/test

Debugging Traefik:

```
docker logs traefik                          # Traefik Logs prüfen
docker ps | grep traefik                     # läuft der Container?
curl -v https://php.m300.smartlearn.ch       # HTTP-Antwort detailliert
docker inspect <containername> | grep -i network  # Netzwerk prüfen
```